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RESEARCH

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Perioperative transcutaneous electrical acupoint stimulation at pericardium 6 reduces postoperative nausea and vomiting after tympanoplasty patients: a randomized controlled study

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Abstract

Background Postoperative nausea and vomiting (PONV) remains a prevalent and distressing complication following tympanoplasty. Stimulation of the Pericardium 6 (PC6) acupoint has emerged as a promising non-pharmacological strategy for PONV management in various surgical procedures. Therefore, this study aimed to evaluate the efficacy of transcutaneous electrical acupoint stimulation (TEAS) at PC6 as an adjunct to standard antiemetic prophylaxis in patients undergoing tympanoplasty.

Methods In this prospective, single-blind, randomized, parallel-group trial, we enrolled 277 patients (aged 18–65 years; ASA physical status I–II) scheduled for tympanoplasty. Participants were randomized to receive either active transcutaneous electrical acupoint stimulation (TEAS group, $n = 138$) or sham TEAS (Sham-TEAS group, $n = 139$). The primary endpoint was the incidence of PONV within 48 h postoperatively. Secondary endpoints included the rate of complete response (CR), the severity of PONV, the requirement for rescue antiemetics, postoperative pain scores, and the incidence of other adverse events.

Results Compared with the Sham group, the TEAS group exhibited a significantly lower incidence of nausea (17.4% vs. 39.1%, $p < 0.01$) and vomiting (6.5% vs. 19.4%, $p < 0.01$). Conversely, the rate of complete response (CR) was significantly higher in the TEAS group than in the Sham group (81.6% vs. 59.7%, $p < 0.01$). No significant intergroup differences were observed regarding recovery parameters or other adverse events.

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Conclusions Perioperative PC6 TEAS serves as a safe and efficacious adjunctive strategy for preventing PONV in patients undergoing tympanoplasty. Our findings support the integration of TEAS into multimodal antiemetic protocols to optimize perioperative care for this high-risk population.

Keywords Postoperative nausea and vomiting, Tympanoplasty, Transcutaneous electrical stimulation, PC6.

Introduction

Postoperative nausea and vomiting (PONV) remains a frequent and distressing complication following general anesthesia, particularly in patients undergoing middle ear surgery [1]. In the absence of prophylaxis, the incidence of PONV can reach as high as 80% within the first 24 h, a phenomenon primarily attributable to the stimulation of the vestibular labyrinth by surgical instrumentation and irrigation fluids [2]. Despite the widespread adoption of multimodal antiemetic strategies, symptom control remains suboptimal; even with dual antiemetic prophylaxis, the residual incidence of PONV has been reported to be as high as 26% [3].

Transcutaneous electrical acupoint stimulation (TEAS) is a contemporary therapeutic approach that integrates principles of transcutaneous electrical nerve stimulation(TENS) with traditional Chinese acupuncture. It involves delivering electrical currents of varying frequencies and intensities via electrode pads positioned over specific acupoints. This non-invasive technique shows potential in reducing the incidence of PONV, accelerating gastrointestinal function recovery, alleviating pain and improving the quality of recovery [4, 5]. Corroborating these findings, a recent meta-analysis of 14 randomized trials(RCTs) confirmed that TEAS can also reduce the incidence of PONV and the need for rescue antiemetics [6]. Although research on TEAS for PONV prophylaxis is increasing, its underlying mechanisms have not been fully elucidated, the underlying physiological mechanisms of TEAS have not yet been fully elucidated. Mechanisms may involve modulation of the autonomic nervous system, reduction of 5-hydroxytryptamine (5-HT) in plasma, inhibition of inflammatory factor release and placebo effects [7, 8].

The Pericardium 6 (PC6) acupoint is currently recognized as a main acupoint of multimodal antiemetic strategies [1]. Stimulation of PC6 via TEAS can reduce the requirement for conventional antiemetic medications while minimizing adverse effects, thereby offering effective PONV prophylaxis. A randomized controlled trial demonstrated that, compared with sham stimulation, PC6 stimulation significantly reduced the incidence of PONV by 12–30% within 72 h following major laparoscopic surgery [9]. Furthermore, our previous study found that PC6 TEAS combined with dexamethasone possesses comparable efficacy to the 5-HT3 antagonist granisetron combined with dexamethasone [10]. However, the efficacy and safety of TEAS stimulation of PC6 in patients for tympanoplasty patients remain unexamined.

This prospective, randomized, evaluator-blinded, parallel-group study was designed to evaluate the efficacy of transcutaneous electrical acupoint stimulation (TEAS) at the Pericardium 6 (PC6) acupoint for the prophylaxis of PONV in patients undergoing tympanoplasty.

Materials and methods

Participants

This single-blind, randomized controlled trial was approved by [Anonymized] and conducted in adherence to the Declaration of Helsinki. The study protocol was prospectively registered at [Anonymized]. Written informed consent was obtained from all participants prior to enrollment.

A total of 277 patients with ASA physical status I–II, aged 18 to 64 years, were enrolled between May and July 2024. Exclusion criteria included: (1) presence of nausea or vomiting, or use of antiemetic medication within 24 h preoperatively; (2) hepatic or renal dysfunction; (3) diabetes or peripheral neuropathy; (4) a history of substance or alcohol abuse; and (5) uncontrolled cardiovascular disease.

Randomization and Blinding Eligible participants were randomized in a 1:1 ratio into two groups using a computer-generated randomization sequence. Allocation was concealed in sequentially numbered, opaque, sealed envelopes, which were opened prior to anesthesia induction. The study employed a single-blind design with blinded outcome assessment: while the attending anesthesiologists performing the intervention could not be blinded, patients, postoperative outcome assessors, ward nurses, and statisticians remained blinded to group allocation throughout the study. To minimize variability, all procedures were performed by a designated team of experienced surgeons and anesthesiologists.

Anesthetic management

A standardized anesthetic protocol was implemented for all participants. Upon arrival in the operating room, standard monitoring was established, including electrocardiography, non-invasive blood pressure, and pulse oximetry. Following 3 min of pre-oxygenation with 100% oxygen, anesthesia was induced with intravenous sufentanil (0.2 μg/kg), propofol (2–4 mg/kg), and rocuronium (0.6 mg/kg). Intravenous dexamethasone (8 mg) was administered immediately post-induction.

Airway management was achieved using an appropriate sized supraglottic airway device. Mechanical ventilation was maintained in pressure-controlled mode, targeting a tidal volume of 6–8 mL/kg and an end-tidal CO_2 ($EtCO_2$) of 35–45 mmHg. Anesthesia maintenance consisted of sevoflurane (0.8–1.0 MAC) and a continuous remifentanil infusion, titrated to maintain mean arterial pressure (MAP) within 20% of baseline values. Depth of anesthesia was monitored via the Bispectral Index (BIS), maintained between 40 and 60. Rocuronium was supplemented as clinically indicated. At wound closure, palonosetron (0.075 mg IV) was administered for antiemetic prophylaxis. To facilitate smooth emergence, a propofol bolus (1 mg/kg IV) was administered immediately prior to the discontinuation of sevoflurane and remifentanil.

In the post-anesthesia care unit (PACU), residual neuromuscular blockade was reversed with sugammadex (2–4 mg/kg) if required. The supraglottic airway was removed upon the return of spontaneous respiration and protective airway reflexes. Recovery metrics, including time to eye opening, command following, and achieving an Aldrete score > 9, were recorded [11].

Transcutaneous electrical acupoint stimulation (TEAS) protocol

In both groups, self-adhesive surface electrodes (Model JNR2; Shanghai Energy Conservation and Environment Protection Technology Co., Ltd., Shanghai, China) were applied bilaterally by an experienced acupuncturist prior to anesthetic induction. The PC6 was located between the tendons of the palmaris longus and flexor carpi radialis muscles, approximately 3 cm proximal to the distal wrist crease [11]. In the TEAS group, electrical stimulation was initiated at 1 mA (2 Hz) and gradually increased (up to 2–20 mA) until the patient experienced a tolerable tingling sensation in the hand. Stimulation was maintained continuously from the induction of anesthesia until the end of surgery. In the Sham-TEAS group, identical electrode placement and procedures were followed to ensure blinding; however, the stimulator was turned off, and no electrical current was delivered.

Outcome assessments

The severity of PONV was assessed using the Simplified PONV Impact Scale (Myles and Wengritzky), with a score of ≥ 5 indicating clinically significant PONV [12]. Rescue antiemetic therapy with droperidol was administered to patients with a scale score ≥ 5, or upon patient request and informed consent for the management of PONV.

Postoperative pain was managed with intravenous oxycodone (0.7 mg/kg) to maintain a numeric rating scale (NRS) score of ≤ 3. Rescue acetaminophen (0.3 g IV) was administered for NRS scores > 4. Complete response (CR) was defined as the absence of nausea and vomiting without the need for rescue antiemetic medication during the study period [13]. To assess the success of blinding, an exploratory survey was conducted during the postoperative follow-up. Patients were asked to guess their group allocation ("active TEAS", "sham TEAS", or "unable to determine").

Data Collection and Outcome Measures—including age, BMI, ASA physical status, Apfel simplified risk score [14], duration of anesthesia and surgery, perioperative opioid consumption, and length of hospital stay—were recorded. The primary outcome was the incidence of nausea and vomiting within the first 48 h postoperatively. Secondary outcomes included the incidence of complete response (CR) within 48 h after surgery, the severity of PONV, use of rescue antiemetic medication, postoperative pain and the occurrence of other adverse events.

Sample size estimation and statistical analysis

The sample size calculation was based on the primary endpoint: the incidence of PONV. Based on a preliminary pilot study conducted with an identical protocol ($n = 10$ per group), the incidence of PONV within 48 h following combined prophylaxis with palonosetron and dexamethasone was 30%. We hypothesized that TEAS would reduce this incidence to 15%. To detect this difference with a power of 80% and a two-sided significance level (α) of 0.05, a minimum of 118 patients per group was required. Assuming a potential dropout rate of 10%, we planned to enroll 132 patients per group (Total $N = 264$).

Data analyses were performed using SPSS software (Version 25.0; IBM Corp., Armonk, NY, USA). The normality of continuous data distribution was assessed using the Kolmogorov-Smirnov test. Normally distributed variables are expressed as mean mean (standard deviation (SD)) and were compared using the Student's t-test. Non-normally distributed variables are presented as median [interquartile range (IQR)] and were analyzed using the Mann-Whitney U test. Categorical data are described as frequencies (percentages) and were compared using the Chi-square test or Fisher's exact test, as appropriate. All statistical tests were two-sided, and a P-value < 0.05 was considered statistically significant.

Results

Patient clinical characteristics

From May to July 2024, a total of 286 patients were assessed for eligibility. Of these, nine were excluded (four did not meet the inclusion criteria; five declined participation), resulting in 277 patients who underwent randomization. Ultimately, all randomized participants (TEAS group, $n = 138$; Sham-TEAS group, $n = 139$) were included in the final analysis (Fig. 1). Baseline

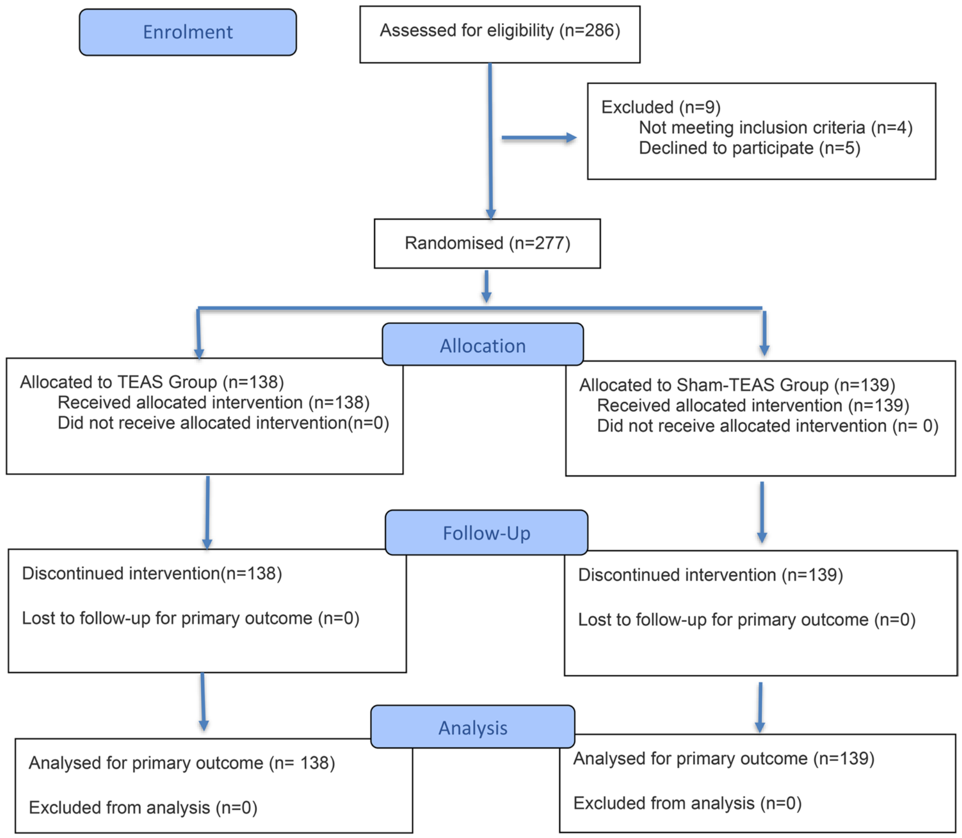

Fig. 1 Flow Diagram

Table 1 Patient characteristics and perioperative data

Variable	TEAS Group (*n* = 138)	Sham-TEAS Group (*n* = 139)	*P*-value
Age (yr)	45.18 (11.93)	46.29(11.86)	0.44
Body mass index (kg.m^{-2})	22.71(3.34)	23.24(3.72)	0.21
ASA physical status (I/II), n	88/50	80/59	0.33
Male/ Female, n	47/91	59/80	0.17
Non-smoker, n	121	111	0.08
History of motion sickness or PONV, n	59	48	0.16
Apfel's score, n			
0/1/2/3/4	15/21/60/42/0	23/29/51/36/0	0.25
Sufentanil equivalents(mg)	15.86(5.11)	15.18(4.95)	0.27
Duration of anesthesia(min)	86.34(30.85)	85.89(28.72)	0.90
Duration of surgery(min)	75.17(29.85)	74.74(28.17)	0.90
Extubation time (min)	18.86(4.21)	19.03(4.75)	0.75
Length of PACU stay (min)	34.98(4.06)	34.19(3.94)	0.10

The data are presented as mean (standard deviation), or number (percentage)

ASA American Society of Anesthesiologists, *PONV* Postoperative nausea and vomiting, *PACU* Post-anesthesia care unit

demographic and perioperative characteristics were well-balanced between the two groups (Table 1).

Incidence of PONV within 48 h postoperatively

As detailed in Table 2, the TEAS group exhibited a significantly lower incidence of vomiting within the 0–48 h postoperative period compared with the Sham-TEAS group (6.52% vs. 19.42%, $P<0.01$). This protective effect was most pronounced in the immediate postoperative phase (0–2 h) (5.07% vs. 12.94%, $P<0.05$). Similarly, the cumulative incidence of nausea within 48 h was significantly reduced in the TEAS group (17.39% vs. 38.13%, $P<0.01$). Notably, the majority of PONV episodes in both groups occurred within the first 24 h. Consequently, the rate of complete response (CR) was significantly higher in the TEAS group than in the Sham-TEAS group (81.16% vs. 59.71%, $P<0.01$).

Severity of PONV

The TEAS group exhibited significantly lower PONV severity scores compared with the Sham-TEAS group during the 0–2, 2–6, and 6–24 h postoperative intervals (all $P<0.05$). However, no statistically significant intergroup differences were observed during the 24–48 h period (Table 3).

Table 2 The incidences of PONV within first postoperative 48 h

Time course	TEAS Group ($n=138$)	Sham-TEAS Group ($n=139$)	P-value
0–2 h			
Nausea	20(14.49)	48(34.53)	< 0.01
Vomiting	7(5.07)	18(12.95)	0.02
2–6 h			
Nausea	7(5.07)	17(12.23)	0.03
Vomiting	4(2.90)	13(9.35)	0.03
6–24 h			
Nausea	0(0.00)	5(3.59)	0.06
Vomiting	0(0.00)	3(2.16)	0.25
24–48 h			
Nausea	0(0)	5(3.59)	0.06
Vomiting	0(0)	2(1.44)	0.25
0–48 h			
Nausea	24(17.39)	53(38.13)	< 0.01
Vomiting	9(6.52)	27(19.42)	< 0.01
Complete response within 48 h after surgery	112(81.16)	83(59.71)	< 0.01

The data are presented as number (percentage)

Rescue antiemetic use

The requirement for rescue antiemetic medication was comparable between the two groups (Table 4). Notably, among patients who experienced PONV, only a subset requested rescue therapy.

Postoperative pain scores

The TEAS group demonstrated significantly lower NRS pain scores at 6 h ($P=0.01$), 24 h ($P<0.01$), and 48 h ($P<0.01$) postoperatively compared with the Sham-TEAS group. In contrast, no significant intergroup difference in pain intensity was observed at the 2-hour time point (Table 5). Regarding rescue analgesia, five patients in each group required acetaminophen for NRS scores > 4, with no significant difference in total analgesic consumption.

Postoperative adverse events

No instances of severe headache, dizziness, or drowsiness—adverse effects commonly associated with palonosetron—were recorded in either group. Furthermore, no adverse cutaneous reactions (e.g., pain or erythema) were observed at the electrode application sites. Overall, no other clinically significant adverse events were reported.

Discussion

This study demonstrates that TEAS at the PC6 significantly reduced the incidence of PONV within the first 48 h postoperatively in patients undergoing tympanoplasty under general anesthesia, compared with sham stimulation. These findings highlight TEAS as a

Table 3 The severity of postoperative vomiting

Time course	TEAS Group ($n=138$)	Sham-TEAS Group ($n=139$)	P-value
0–2 h			< 0.01
0	116(84.06)	91(65.47)	
1	11(8.00)	22(15.83)	
2	7(5.07)	17(12.23)	
3	3(2.17)	7(5.04)	
4	1(0.72)	1(0.72)	
5	0(0.00)	1(0.72)	
6	0(0.00)	0(0.00)	
2–6 h			0.02
0	130(94.20)	119(85.61)	
1	4(2.90)	12(8.63)	
2	3(2.20)	4(2.88)	
3	1(0.72)	2(1.44)	
4	0(0.00)	1(0.72)	
5	0(0.00)	1(0.72)	
6	0(0.00)	0(0.00)	
6–24 h			0.03
0	138(100.00)	134(96.40)	
1	0(0.00)	2(1.44)	
2	0(0.00)	1(0.72)	
3	0(0.00)	1(0.72)	
4	0(0.00)	0(0.00)	
5	0(0.00)	1(0.72)	
6	0(0.00)	0(0.00)	
24–48 h			0.08
0	138(100.00)	136(97.84)	
1	0(0.00)	1(0.72)	
2	0(0.00)	1(0.72)	
3	0(0.00)	0(0.00)	
4	0(0.00)	1(0.72)	
5	0(0.00)	0(0.00)	
6	0(0.00)	0(0.00)	

The data are presented as number (percentage)

Table 4 The consumption of rescue antiemetics during postoperative 48 h

Time course	TEAS Group ($n=138$)	Sham-TEAS Group ($n=139$)	P-value
0–2 h	7 (5.07%)	5(3.60%)	0.55
2–6 h	1 (0.72%)	1 (0.72%)	1.00
6–24 h	0(0.00)	1 (0.72%)	1.00
24–48 h	0(0.00)	0(0.00)	1.00
0–48 h	0(0.00)	0(0.00)	1.00

The data are presented as number (percentage)

Table 5 Postoperative pain during postoperative 48 h

Time course	TEAS Group ($n=138$)	Sham-TEAS Group ($n=139$)	P-value
Postoperative 2 h	1.67 (0.82)	1.66(0.93)	0.91
Postoperative 6 h	1.55 (0.85)	1.86 (0.97)	0.01
Postoperative 24 h	1.30(0.92)	1.65(0.95)	< 0.01
Postoperative 48 h	0.82(0.75)	1.09(0.75)	< 0.01

The data are presented as mean (standard deviation) promising, non-pharmacological adjunct for the management of PONV in this population, effectively complementing existing pharmacological strategies.

Middle ear surgeries are associated with a high risk of postoperative nausea and vomiting (PONV), with incidences reaching 80% in patients devoid of prophylactic antiemetic treatment [2]. This elevated susceptibility is attributed to a multifactorial etiology. First, surgical maneuvers such as drilling and suctioning generate low-frequency vibrations within the mastoid cavity; these vibrations may disrupt the vestibular system and induce postural instability, thereby triggering PONV. Second, intraoperative irrigation fluids may directly stimulate the vestibular labyrinth via the vestibular division of the vestibulocochlear nerve (CN VIII). This afferent stimulation activates the chemoreceptor trigger zone (CTZ), thereby precipitating nausea and vomiting. Furthermore, intraoperative anesthetic agents and analgesics contribute to this process by directly stimulating the CTZ. Similarly, circulating serotonin (5-HT) may bind to 5-HT$_3$ receptors in this region, further triggering the emetic reflex [3, 15, 16].

Current guidelines recommend multimodal antiemetic prophylaxis for patients at moderate to high risk of PONV. This strategy typically involves combining dexamethasone with 5-HT$_3$ receptor antagonists, neurokinin-1 (NK-1) receptor antagonists, or dopamine receptor antagonists [1]. Palonosetron, a second-generation 5-HT$_3$ receptor antagonist, is distinguished by its unique allosteric inhibition mechanism, which confers greater potency and a prolonged half-life compared to other agents in its class. It exhibits a receptor binding affinity approximately 30-fold higher than that of ondansetron or ramosetron, with an effective duration of action approaching 40 h [17]. Previous studies have established 0.075 mg as the minimum effective dose of palonosetron for reducing the severity of nausea within the first 24 h postoperatively, as well as for delaying the onset of emesis and treatment failure [18].

Despite palonosetron's superior efficacy over other antiemetics in middle ear surgery [19], the incidence of PONV remains substantial, even when combined with dexamethasone [3]. Consequently, concerns regarding treatment costs and potential adverse effects have prompted growing interest in non-pharmacological interventions, which are increasingly recommended as adjunctive strategies in current clinical guidelines.

TEAS is a non-invasive, non-pharmacological intervention that delivers electrical stimulation to specific acupoints via skin electrodes. Unlike traditional acupuncture, TEAS is particularly well-suited for the perioperative setting due to its non-invasiveness, ease of administration, and high acceptability among both patients and surgeons [20]. Existing literature supports the utility of TEAS as a multimodal antiemetic adjunct, demonstrating its efficacy in reducing PONV incidence in various surgical populations, including those undergoing laparoscopic procedures [21]. Furthermore, TEAS has been shown to decrease intraoperative analgesic requirements and mitigate anesthesia-related complications [22]. In our previous work, we found that PC6 stimulation combined with dexamethasone achieved efficacy comparable to that of the 5-HT$_3$ antagonist-dexamethasone combination in preventing PONV in breast cancer patients [10]. This regimen suggests potential advantages regarding cost-effectiveness and a favorable safety profile. However, the efficacy and safety of TEAS specifically in patients undergoing tympanoplasty remain investigating. Moreover, the underlying mechanisms of TEAS are not fully elucidated but may involve neuroendocrine modulation—specifically, the release of endogenous opioids and the activation of adrenergic pathways, which in turn alter 5-HT transmission to inhibit the emetic reflex [23].

In this study, PC6 was selected as the stimulation site given its established status as a primary locus for PONV prophylaxis [24]. Mechanistically, PC6 stimulation is hypothesized to exert antiemetic effects by modulating neuroendocrine pathways and inducing homeostatic autoregulation. Furthermore, existing evidence indicates that acupuncture at PC6 enhances vagal tone, which plays a pivotal role in suppressing the emetic reflex [25]. However, the extent to which these physiological mechanisms translate to TEAS intervention specifically within the tympanoplasty population remains to be verified.

Notably, previous research suggests that PC6 stimulation may not only prevent PONV but also attenuate postoperative pain. While the underlying mechanisms remain to be fully elucidated, they likely involve the modulation of the endogenous opioid system, including the release of endorphins, the regulation of neurotransmitters such as 5-HT, norepinephrine, and substance P, and the modulation of the endogenous opioid system [26]. In the present study, no significant difference in pain scores was observed between the two groups during the initial 0–2 h postoperatively, a finding potentially attributable to the residual effects of intraoperative analgesics. However, from 2 to 48 h postoperatively, the TEAS group exhibited significantly lower pain scores, suggesting that PC6 TEAS provides effective analgesia during the intermediate postoperative period. These findings align with previous studies demonstrating the analgesic efficacy of TEAS in gastric and colorectal surgeries [27, 28], as well as in the management of chronic pain following mastectomy [29], Furthermore, TEAS has been associated with broader benefits, such as reducing postoperative ileus and enhancing gastrointestinal recovery [30].

In contrast to previous studies demonstrating reduced rescue antiemetic use with PC6 stimulation [31, 32], we observed no significant difference in rescue medication consumption between groups. This discrepancy may be attributed to two factors. first, First, our protocol enforced strict administration criteria based on the Simplified PONV Impact Scale [12],, restricting rescue medication to cases of clinically significant PONV (severity score ≥ 5). A post-hoc analysis (Table 3) revealed that severe PONV was extremely rare (0.72% in Sham-TEAS vs. 0% in TEAS); instead, TEAS predominantly mitigated mild-to-moderate episodes (score < 5) that remained below the threshold for rescue intervention. Second, the requirement for patient consent prior to administration meant that some patients, concerned about potential side effects, may have declined rescue therapy, potentially masking intergroup differences. Collectively, while TEAS significantly reduced overall PONV incidence and severity (Tables 2 and 3), these findings suggest its clinical value lies primarily in alleviating the patient's subjective burden of PONV rather than altering the consumption of rescue antiemetics, a conclusion consistent with findings in craniotomy patients [33].

Our study has several limitations. First, although a single-blind design was employed, maintaining strict blinding was challenging due to the distinct somatosensory sensations associated with active stimulation. The proportion of patients correctly guessing their allocation was 77.5% in the TEAS group and 66.2% in the sham group, both significantly exceeding chance (50%). Consequently, we cannot entirely rule out the contribution of nonspecific psychological effects to the observed outcomes. Second, follow-up was limited to 48 h postoperatively. While this window captures the peak incidence of PONV, it precludes the assessment of delayed recovery or long-term recurrence. Third, as a single-center trial restricted to ASA I–II patients aged 18–64 undergoing tympanoplasty, the external validity of our findings is limited. The generalizability of these results to high-risk populations (ASA III–IV), other surgical procedures, or alternative anesthetic protocols remains uncertain. Future multicenter studies with stratified sampling are warranted to validate the broader clinical applicability of TEAS.

Conclusion

In conclusion, compared with conventional 5-HT receptor antagonists combined with dexamethasone, PC6 TEAS demonstrated a trend toward improved efficacy in reducing both PONV incidence and perioperative antiemetic consumption in patients undergoing tympanoplasty. These preliminary findings suggest that PC6 TEAS may serve as a feasible, non-pharmacological adjunct to multimodal antiemetic regimens for this population. Integrating TEAS into standard care protocols could potentially optimize antiemetic outcomes, minimize reliance on pharmacotherapy, and thereby reduce associated medical costs and adverse events.

Abbreviations

PONV	Postoperative nausea and vomiting
PC6	Pericardium 6 acupuncture point
TEAS	Transcutaneous electrical acupoint stimulation
CR	Complete response
ASA	American Society of Anesthesiologists physical status
$EtCO_2$	End-tidal CO_2
MAC	Minimum alveolar concentration
MAP	Mean arterial pressure
BIS	Bispectral index
PACU	Postanaesthesia care unit
POV	Postoperative vomiting
NRS	Numeric rating scale
CTZ	Chemoreceptor trigger zone
NK-1	Neurokinin-1

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12906-026-05357-8.

Supplementary Material 1.

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Not applicable.

Authors' contributions

Conception and design of the research: Jie Jia. Acquisition of data: Shuangshuang Li, Yaojing Fu. Analysis and interpretation of the data: Lu Luo, Jie Jia. Statistical analysis: Lu Luo, Yaojing Fu, Shuangshuang Li. Obtaining financing: None. Writing of the manuscript: Lu Luo, Yaojing Fu. Critical revision of the manuscript for intellectual content: Jie Jia.

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Data availability

No datasets were generated or analyzed during the current study. However, the data and materials are available from Jie Jia upon reasonable request. Please contact Jie Jia for further information regarding data availability.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the Eye & ENT Hospital of Fudan University (Approval number: 2024041) and informed consent was obtained from all participants.

Consent for publication

The authors declare that the manuscript has not been published and is not currently being considered for publication in another journal.

Competing interests

The authors declare no competing interests.

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